Synthetic e-commerce fulfillment network

A three-part dataset for network design and order-routing work: customer orders, the facilities that can ship them, and what it costs and how long it takes to get from each facility to each destination.

12,000	10	300	297
orders over 90 days	facility nodes	cost and lead-time pairs	total daily order capacity

How the three tables fit together

Each order is shipped to a ZIP code. The first three digits (ZIP3) key into the cost and lead-time matrices, which hold one value for every facility and ZIP3 pair. A facility can fill an order only if the heaviest SKU in it is within the facility's weight limit, and only while it has daily order capacity left. Choosing which facility fills each order, under those limits, is the decision this data is built to support.

Table	Rows	Key fields	Role
orders	12,000	order_id, timestamp, zip_code, basket_size, sku_weight_lb, total_weight_lb	Demand: what is ordered, when, and where it goes
facility_nodes	10	node_id, type, lat, lon, daily_order_capacity, storage_capacity_units, max_sku_weight_lb, fixed_cost_per_day_usd	Supply: where orders can ship from and how much each node can handle
transport_cost_usd	10 x 30	node_id by ZIP3	Shipping cost per order for each facility and destination
lead_time_days	10 x 30	node_id by ZIP3	Days from facility to destination

What is in the data

- Orders run from 2026-07-01 to 2026-09-28, averaging 133 a day. Daily volume rises from about 117 in the first two weeks to about 139 in the last two, with spikes on two sale events: July 15-16 and September 5-7. The peak day, September 5, has 230 orders.
- Orders arrive in the evening: the busiest hour is 20:00, and weekends run about 20% above weekdays.
- Baskets average 2.7 items. Most SKUs are light (median 1.5 lb), but 7% of orders contain an oversize SKU over 20 lb, up to 80 lb. Retail stores cap SKU weight at 40 lb, so the heaviest items in that range can ship only from warehouses.
- Orders go to 30 metro areas (2,761 distinct 5-digit ZIP codes). New York, Los Angeles and Chicago account for about 31% of volume.
- Four warehouses carry most of the capacity. Six retail stores add small, high-cost, last-mile capacity in dense metros.

All data is synthetic, generated with a fixed random seed so it can be reproduced. It does not describe any real company, customer or shipment.

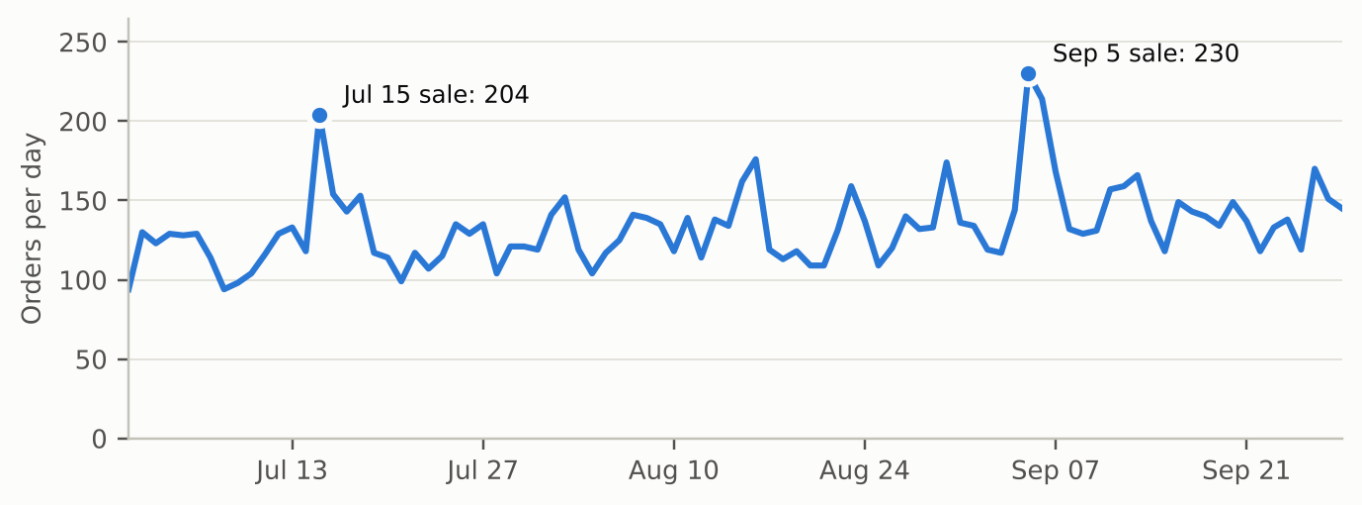
1. Orders

12,000 synthetic orders. Each row is one order with a single heaviest-SKU weight, which is what limits which facilities can fill it.

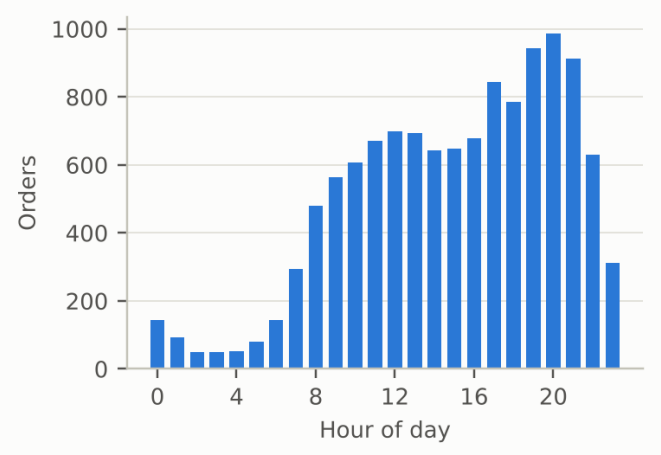
Field	Type	Description
order_id	text	Unique order number, ORD-000001 onward
timestamp	datetime	When the order was placed
zip_code	text	5-digit destination ZIP; first 3 digits (ZIP3) key the cost matrices
basket_size	integer	Items in the order, 1 to 10
sku_weight_lb	decimal	Weight of the heaviest SKU in the order, pounds
total_weight_lb	decimal	basket_size x sku_weight_lb, an upper-bound estimate of shipment weight

Measure	Mean	Median	95th pct	Max
Orders per day	133	-	-	230
Basket size (items)	2.7	3	5	10
SKU weight (lb)	5.1	1.5	36.0	79.9
Total order weight (lb)	13.9	3.7	72.1	583

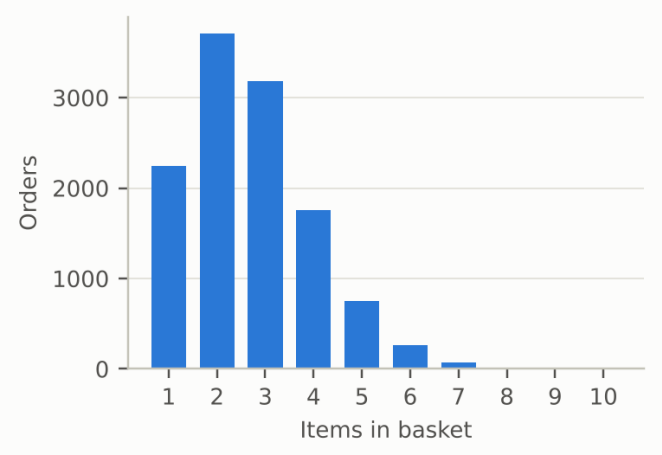
Orders per day



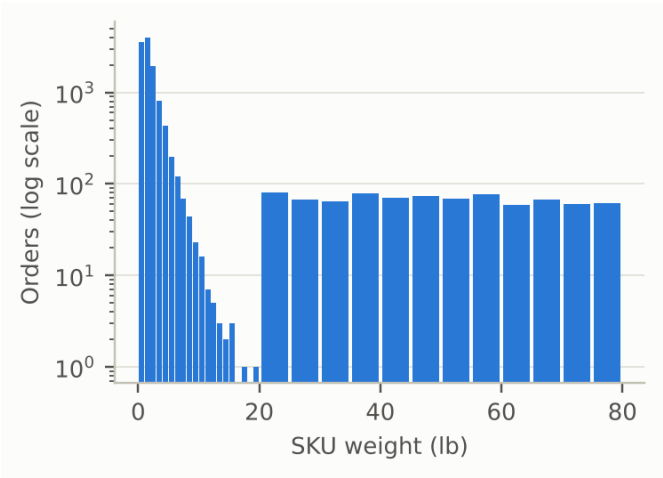
Orders by hour of day



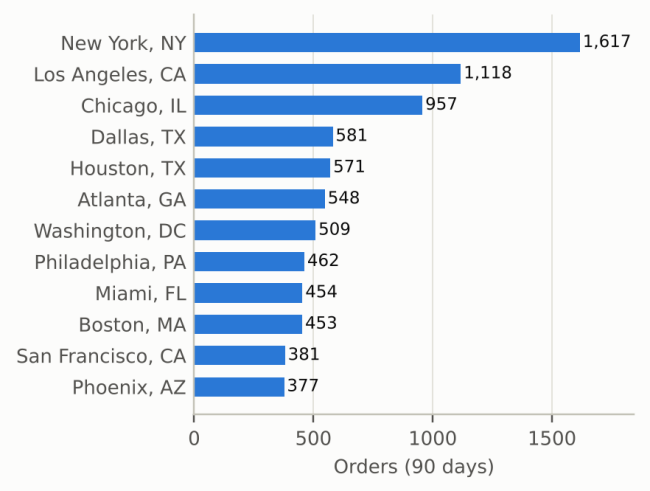
Basket size



Heaviest-SKU weight (1 lb bins to 20 lb, 5 lb bins above; log scale)



Top 12 destination metros



Sample rows

order_id	timestamp	zip_code	basket_size	sku_weight_lb	total_weight_lb
ORD-000001	2026-07-01 01:32:37	28249	1	5.41	5.41
ORD-000002	2026-07-01 08:35:06	63135	3	4.44	13.32
ORD-000003	2026-07-01 08:37:56	78794	4	6.96	27.84
ORD-000004	2026-07-01 08:51:15	75237	4	1.00	4.00
ORD-000005	2026-07-01 09:06:34	97205	3	0.69	2.07
ORD-000006	2026-07-01 09:07:38	98112	2	44.65	89.30
ORD-000007	2026-07-01 09:08:46	80251	1	1.46	1.46
ORD-000008	2026-07-01 09:17:44	77003	2	0.87	1.74
...	...	...	...	...	...
ORD-000006	2026-07-01 09:07:38	98112	2	44.65	89.30
ORD-000009	2026-07-01 09:35:39	37277	3	74.11	222.33
ORD-000017	2026-07-01 11:09:19	97213	2	74.49	148.98

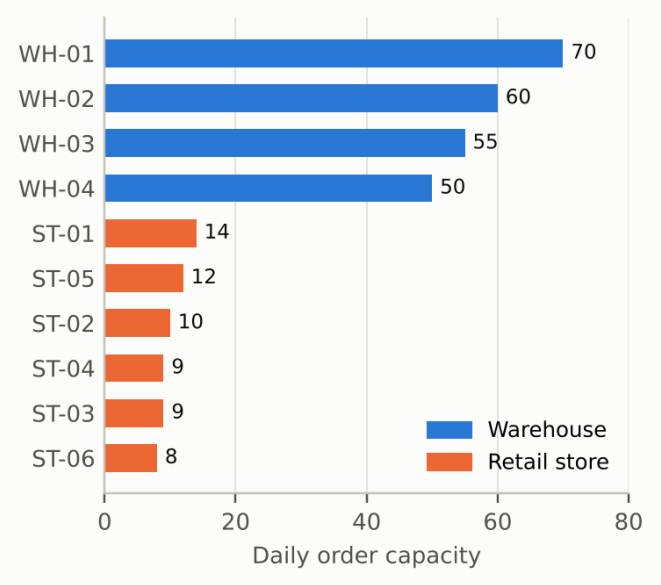
The last three rows are oversize orders (SKU over 20 lb). Stores can fill them only if the SKU is under their 40 lb limit.

2. Facility nodes

Four warehouses and six retail stores. Warehouses handle the volume and any weight. Stores are cheaper to reach in their own metro but have small order capacity, small storage and a 40 lb SKU weight limit.

Node	Name	Type	Location	Orders / day	Storage (units)	Max SKU (lb)	Fixed cost / day
WH-01	Edison Fulfillment Center	Warehouse	Edison, NJ	70	90,000	150	$1,850
WH-02	Joliet Fulfillment Center	Warehouse	Joliet, IL	60	80,000	150	$1,600
WH-03	Grand Prairie Fulfillment Center	Warehouse	Grand Prairie, TX	55	75,000	150	$1,500
WH-04	Ontario Fulfillment Center	Warehouse	Ontario, CA	50	70,000	150	$1,700
ST-01	Atlanta Retail Store	Retail store	Atlanta, GA	14	6,500	40	$240
ST-02	Miami Retail Store	Retail store	Miami, FL	10	5,200	40	$210
ST-03	Denver Retail Store	Retail store	Denver, CO	9	4,800	40	$200
ST-04	Seattle Retail Store	Retail store	Seattle, WA	9	4,600	40	$230
ST-05	Boston Retail Store	Retail store	Boston, MA	12	5,800	40	$250
ST-06	Phoenix Retail Store	Retail store	Phoenix, AZ	8	4,200	40	$190

Daily order capacity by node



Reading the table

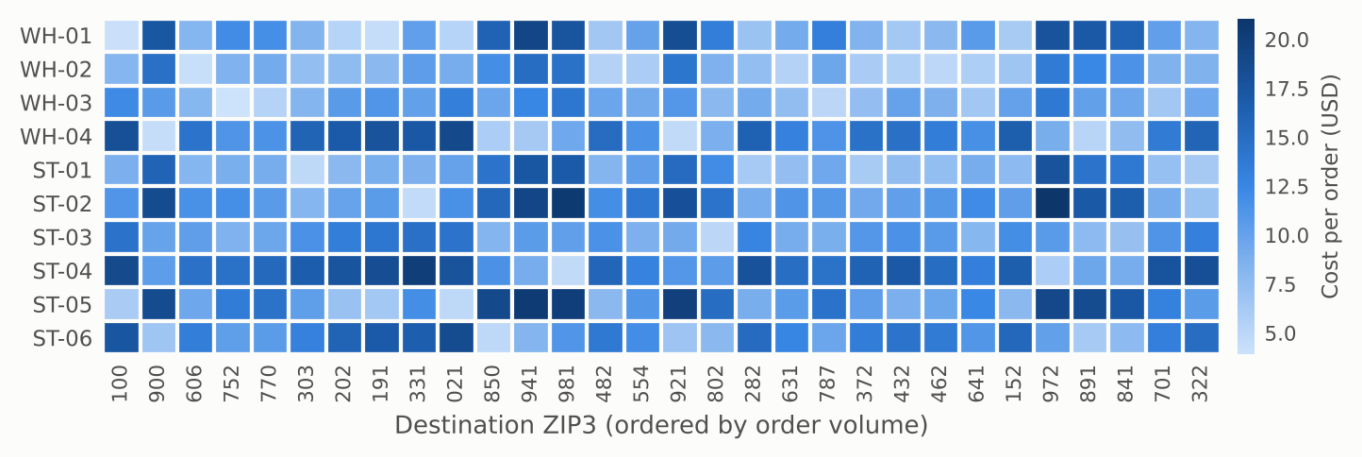
Warehouses hold 50 to 70 orders per day each and 70,000 to 90,000 storage units, so any one of them can absorb a day's worth of regional demand. A store holds 8 to 14 orders per day. When a metro has a store, the store is the cheapest option for light orders there, which is why stores run near or over capacity on busy days.

Capacity is stated per day. Fixed cost is the daily cost of keeping a node open, and it is separate from per-order shipping cost.

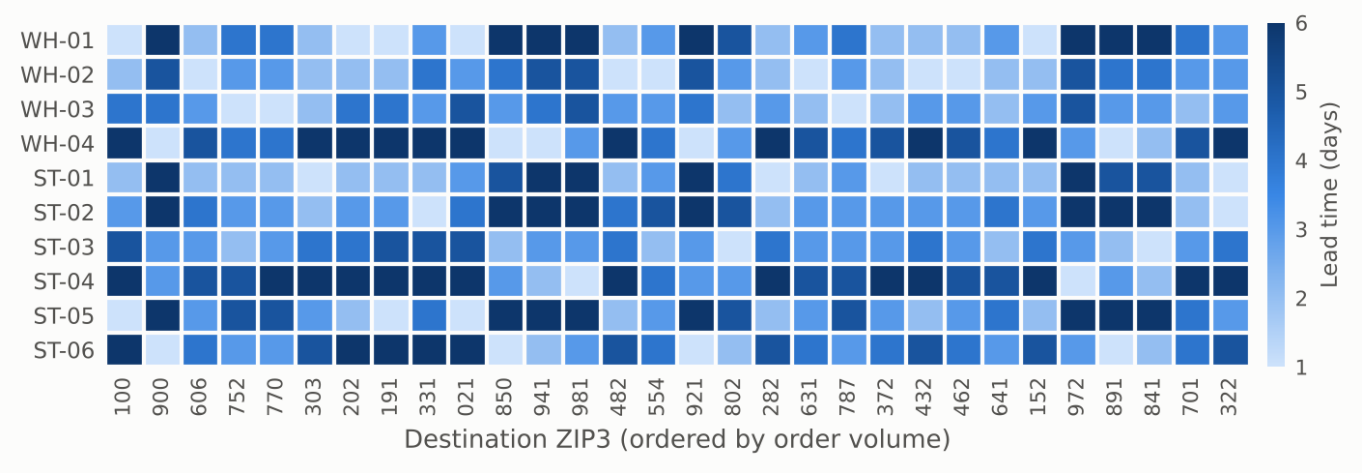
3. Transportation cost and lead-time matrices

Two matrices, each with 10 facilities as rows and 30 destination ZIP3 areas as columns. Cost is dollars per order. Lead time is whole days from facility to destination. Both grow with road distance, which is great-circle distance scaled by 1.2.

Cost per order (USD)



Lead time (days)



Excerpt: the eight highest-volume destinations

Node	100 New York	900 Los Angeles	606 Chicago	752 Dallas	770 Houston	303 Atlanta	202 Washington	191 Philadelphia
WH-01	$4.08 / 1d	$17.31 / 6d	$8.17 / 2d	$11.95 / 4d	$11.73 / 4d	$8.38 / 2d	$5.28 / 1d	$4.37 / 1d
WH-02	$8.14 / 2d	$14.72 / 5d	$4.17 / 1d	$8.55 / 3d	$9.15 / 3d	$7.42 / 2d	$7.64 / 2d	$7.87 / 2d
WH-03	$12.18 / 4d	$10.70 / 4d	$8.12 / 3d	$3.90 / 1d	$5.34 / 1d	$8.21 / 2d	$10.65 / 4d	$11.21 / 4d
WH-04	$18.11 / 6d	$4.39 / 1d	$14.40 / 5d	$11.22 / 4d	$11.38 / 4d	$16.03 / 6d	$16.94 / 6d	$17.74 / 6d
ST-01	$8.78 / 2d	$15.95 / 6d	$8.16 / 2d	$8.84 / 2d	$9.06 / 2d	$4.83 / 1d	$7.96 / 2d	$8.85 / 2d
ST-02	$11.20 / 3d	$18.61 / 6d	$11.57 / 4d	$11.70 / 3d	$10.66 / 3d	$8.17 / 2d	$9.91 / 3d	$10.55 / 3d
ST-03	$14.46 / 5d	$9.90 / 3d	$10.30 / 3d	$8.58 / 2d	$9.56 / 3d	$11.50 / 4d	$13.35 / 4d	$13.99 / 5d
ST-04	$18.76 / 6d	$10.41 / 3d	$14.64 / 5d	$14.61 / 5d	$15.41 / 6d	$16.59 / 6d	$17.61 / 6d	$18.38 / 6d
ST-05	$6.05 / 1d	$18.64 / 6d	$9.52 / 3d	$13.54 / 5d	$14.52 / 5d	$10.38 / 3d	$7.01 / 2d	$6.42 / 1d
ST-06	$17.33 / 6d	$6.68 / 1d	$13.34 / 4d	$10.38 / 3d	$10.56 / 3d	$13.05 / 5d	$16.03 / 6d	$16.86 / 6d

Each cell is cost per order and lead time in days.

4. A baseline to beat

As a first reference, every order was sent to the cheapest facility that can take its heaviest SKU, ignoring capacity. This is the naive policy a routing model should improve on.

Node	Share of orders	Avg daily load / capacity	Days over capacity (of 90)
WH-01	23.8%	45%	0
WH-02	22.0%	49%	0
WH-03	12.9%	31%	0
WH-04	16.4%	44%	0
ST-01	7.2%	69%	9
ST-02	3.7%	49%	2
ST-03	3.2%	48%	1
ST-04	4.3%	64%	8
ST-05	3.6%	40%	1
ST-06	3.0%	50%	1

Under this policy the average order costs $4.93 to ship with an average lead time of 1.06 days. Total capacity is not the problem: warehouses average 31% to 49% utilization. The problem is local. Stores in Atlanta and Seattle exceed their daily capacity on 9 and 8 days, and 61 orders (0.5%) exceed what a node can handle in a day. A capacity-aware routing model would push that overflow to warehouses at a small extra shipping cost.

How the data was generated

- **Orders.** Daily volume follows a weekday pattern with gradual growth over 90 days and sale-event spikes. Hour of day follows an evening-peaked curve. Destinations are drawn from 30 metro areas weighted by population, with a random 2-digit suffix to make a 5-digit ZIP. Basket size is 1 plus a Poisson draw, capped at 10. SKU weight is log-normal around 1.4 lb, with 7% of orders drawn from a 20 to 80 lb oversize range.
- **Facilities.** Four warehouse sites and six store sites placed in or near the demand metros. Capacities, weight limits and fixed costs are set by hand.
- **Matrices.** Cost per order is $4.00 plus $0.0048 per road mile, plus $0.80 for store origins, with 3% random noise. Lead time is 1 day plus 1 day for each 450 road miles, capped at 6.

The figures in this brief are computed from the generated tables. The generator uses a fixed random seed, so the same data can be regenerated exactly.